CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

M.Tech. (CAD/CAM) **Sem - I** **University Examination, 2009-10**

CE – 725 Artificial Intelligence

Date: 07.01.11, Tuesday Time: 10:30 a.m To 01:30 p.m Maximum Marks: 70

Instructions: (1) All questions are compulsory.
(2) Indicate clearly, the option you attempt along with its respective question number.
(3) Figures to right indicate marks
(4) Rough work is done in the last page of main answer book, don't write anything on the question paper.

SECTION-I

Q-1 **Answer the following questions.**

1. Explain search space and problem space with the help of a suitable example. **3**
2. Discuss the important characteristics of production system and control strategy. **3**
3. What are the disadvantages of Hill Climbing Search Technique? Mention the possible solution(s) for each problem. **2**
4. Which is better when the memory is limited, Breadth First Search or Depth First Search? Give Reason(s). **3**

Q-2

[A] Give an example of problems for which BFS would work better than DFS. Explain it in detail. **4**

[B] Write a PROLOG program for reversing a list using difference list concept. **4**

[C] Draw the semantic net for following sentence: "Every dog in the town has bitten the constable". **4**

OR

[A] Explain: Multilayer Feed forward Networks. **4**

[B] Draw the block diagram of a typical Expert System. Explain in brief each component of it. **4**

[C] How AND Gate problem can be solved by Neural Network. Explain in brief. **4**

Q-3

[A] Convert these sentences to predicate logic. Using the logical rules, proof by resolution that prove "whether John eats Kumquats or not". **4**
(a) John likes fruit.
(b) All fruits are not liked by John.
(c) Kumquats are fruits.
(d) John eats only those fruits, which he likes.

[B] Show how means-ends analysis could be used to solve the problem of Robotics. **4**

[C] Explain the key idea of Bayesian statistics for representing knowledge. Is it possible to compute P (A/~B)? Explain. **4**

OR

[A] Write a PROLOG program to find factorial of a given number and to generate the Fibonacci series. **4**

[B] Explain Back propagation Algorithm. What is Back propagation error? **4**

[C] Write a PROLOG program to generate family tree. **4**

SECTION-II

Q-4

1. Explain the term syntactic processing and semantic analysis in context to NLP. **3**
2. Compare Nonhierarchical Planning with Hierarchical Planning. **4**
3. What is a Clausal form? How it is related with PROLOG? Why PROLOG is called declarative language? **4**

Q-5

[A] Write down the water jug problem. Give the rules for water jug problem. **4**

[B] Compare Fuzzy Logic with other Logics. Give the real life example where Fuzzy logic concept is used. **4**

[C] Give the significance of the following terms with respect to Genetic Algorithm. Crossover, Mutation, Chromosome and Generation. **4**

OR

[A] Describe four major problems faced by an Expert System. **4**

[B] Give various features of Perceptron network. **4**

[C] A problem solving search can proceed either in forward or backward. Whether search should proceed forward or backward for the following problems. Justify the answer with the reason. **4**

1. Planning a party.
2. Going back to home from unknown place.

Q-6

[A] Derive the parse tree for the sentence using the appropriate rules: **4**
John gave the book to marry.

[B] Explain the working of Dempster Shafer theory with an example. Also show the use of Plausibility in it. **4**

OR

[B] Solve the following crypt arithmetic problem. **4**

$$\begin{array}{r} \textit{SEND} \\ \underline{+\textit{MORE}} \\ \textit{MONEY} \end{array}$$

[C] Differentiate: Supervised and Unsupervised learning. Give example of both. **4**

OR

[C] Give the practical situations where cur, fail, negation and recursion may be useful in PROLOG. **4**

$$\begin{array}{r} 9567 \\ \underline{+\ 1085} \\ 10652 \end{array}$$